



Refrigerator Volume & Thermal Cycle Calculator – User Guide

Welcome to the Refrigerator Conceptual Design Tool. This application allows engineers to quickly prototype cabinet geometries, compute precise internal volumes, and run thermodynamic cycle simulations to balance heat loads against compressor capacities.

1. Volume Calculation

The **Volume** tab focuses entirely on the spatial and structural properties of the cabinet.

Defining Cabinet Geometry

1. **External Dimensions:** Enter the overall Height (H), Width (W), and Depth (D) of the cabinet in millimeters.
2. **Compressor Step (Hb, Db1, Db2):** Define the stepped floor at the bottom rear of the cabinet where the compressor sits.
3. **Wall Thicknesses:** Specify the polyurethane (PU) insulation thickness for all exterior walls and the dividing partition.

Managing Compartments

- Choose between a single-door (1 compartment) or double-door (2 compartments) configuration.
- Assign each compartment as either **Freezer** or **Fresh**.
- Use the **Ratio (%)** or **Height (mm)** fields to adjust how the internal vertical space is distributed.
- Specify the **Number of Shelves**. The system will automatically calculate the volumetric deduction of the shelf rails based on your settings.

Obstacles & Net Volume

The tool calculates **Gross Volume** and **Total Usable Volume**. To get accurate usable volume, define the dimensions of internal obstacles:

- **Evaporator Depth:** The space lost at the back of the freezer.
- **Control Box & R-Shower:** Space lost in the fresh food compartment.
- **Door Dikes:** The volume consumed by the door interior liners protruding into the cabinet.

Click **Calculate** to update the 2D visual schematics and generate the volume report.

2. Thermal Analysis

The **Thermal** and **Inverter** tabs simulate the refrigeration cycle based on the active cabinet geometry.

Running a Simulation

- 1. **Target Temperatures:** Set the desired ambient (T_0), freezer (T_F), and refrigerator (T_R) temperatures.
- 2. **Refrigerant:** Select the system refrigerant (e.g., R-600a, R-134a).
- 3. **Running Ratio (PR):** For inverter compressors, specify the target running ratio (e.g., 0.85). The solver will calculate the required RPM to maintain this ratio.
- 4. Click **Run Thermal Analysis**. The system will calculate the heat leak through the cabinet walls, resolve the Evaporator (T_E) and Condenser (T_C) temperatures, and output the required compressor capacity, COP, and energy consumption.

Advanced Thermal Parameters

Click the  **Advanced** button inside the Thermal tab to fine-tune the system:

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